Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

4th Year 1st Semester Final Examination, 2024

Course Code: EEE 459, Session: 2020-2021

Course Title: Compound Semiconductor and Hetero-Junction Devices

Full Marks: 100 **Credit: 3.0** **Total Time: 3 Hours**

Part-A

*[Answer any **two** of the following questions]*

1. (a) Between Si and GaAs, which has higher electron mobility and why? 6

   (b) Discuss Velocity saturation and Gunn effect . 5

   (c) Prove that along the [100] and [111] direction in k-space, for Si, the variation of electron energy with respect to wave number is higher for light holes than for heavy holes. 6

   (d) For a $GaAs - Al_xGa_{1-x}As$ heterojunction, determine the following:

   (i) Energy-Band diagram: Case-1: Before contact, Case-2: After contact

   (ii) Show that the built-in potential (using hole concentration) $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{P_P\, N_{VN}}{P_N\, N_{VP}}\right)$. Given, GaAs is doped P type, $Al_xGa_{1-x}As$ is doped N^+ type. Assume: $x = 0.25$ & $\Delta E_c = \frac{1}{2}\Delta E_g$ P_p = hole concentration in P type region, P_N = hole concentration in N type region, N_{VN} and N_{VP} are density of states of electron and hole respectively. 8

2. (a) Which semiconductors are most extensively used in near infra-red region? Explain the reasons behind it. 5

   (b) Why is lattice matching of semiconductor important for hetero-structured devices? Discuss the effect of strain and critical thickness in epitaxial growth. 9

   (c) Discuss the effect of isoelectric trap in the ternary compound semiconductor $GaAs_{1-x}P_x$. 4

   (d) Fig. 1 shows the Bandgap Vs. lattice constant for different binary compounds. Assume that a quaternary compound $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. Find the composition if the bandgap is 1 eV and lattice parameter is 5.93 Å. 7

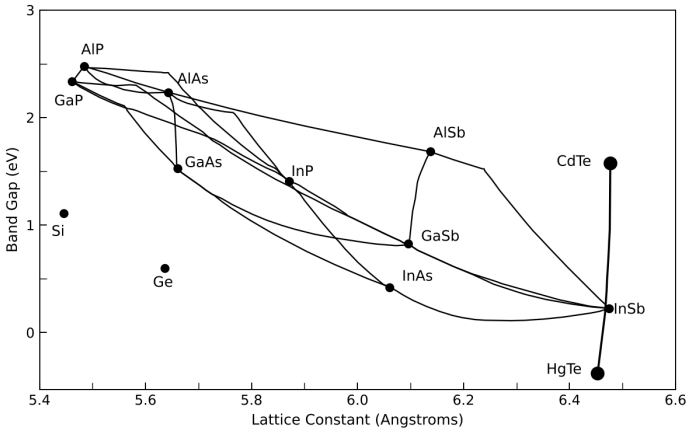


Fig. 1: Band gap vs. lattice constant of binary compounds (redrawn from the paper)

3. (a) Draw the ideal energy band diagram of a metal semiconductor junction under:

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(i) Reverse bias condition
(ii) Forward bias condition 5

(b) Explain how the density of states changes qualitatively as the dimensionality reduces from $3D \rightarrow 2D \rightarrow 1D$. Sketch $D(E)$ vs E for each case. 6

(c) Consider a contact between tungsten and n-type silicon doped to $N_d = 1 \times 10^{16}$ cm^{-3} at $T = 300$ K. Determine the

(i) theoretical barrier height
(ii) built-in potential barrier
(iii) maximum electric field in the metal-semiconductor diode for zero applied bias, and
(iv) the junction capacitance. Given that, the metal work function for tungsten (W) is 4.55 V and the electron affinity for silicon is 4.01 V. Assume $N_c = 2.8 \times 10^{19}$ cm^{-3}. 8

(d) Consider a gallium arsenide metal-semiconductor contact in which the electric field in the semiconductor is assumed to be $E = 6.8 \times 10^4$ V cm^{-1}. Calculate the Schottky barrier lowering (due to image force induced lowering) and the position of the maximum barrier height. 6

Part-B

*[Answer any **two** of the following questions]*

4. (a) A narrow bandgap N-type and a wide bandgap P-type material are used to form a heterojunction. If they are straddling type, then draw the energy band diagram of the heterojunction before contact and after contact at thermal equilibrium. With necessary assumption, derive expression for electric field and electrostatic potential in both regions. 12

(b) Discuss the following non-ideal effects on barrier height of a metal-semiconductor heterojunction:

(i) Image-force-induced lowering
(ii) Interface states 8

(c) Describe how a linearly graded junction is formed. 5

5. (a) Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where V_p, V_g, and V_d are pinch-off, gate, and drain voltage respectively)

(i) $V_g > V_p$ and $V_d < V_{d,sat}$
(ii) $V_g > V_p$ and $V_d = V_{d,sat}$
(iii) $V_g > V_p$ and $V_d > V_{d,sat}$ 6

(b) Derive the formula of internal pinch-off voltage in an n-channel JFET. 8

(c) Consider a GaAs n-channel pn JFET at $T = 300$ K with $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 3 \times 10^{15}$ cm^{-3}, $\epsilon_s = 13.1$, and $a = 0.70$ μm. Determine the forward-bias gate voltage required to open a channel region that is 0.10 μm thick with zero drain voltage. 7

(d) Sketch the ideal energy band diagram of a metal semiconductor junction in which $\Phi_m < \Phi_s$. Explain why this is an ohmic contact. 4

6. (a) Draw the equivalent circuit for basic Ebers-Moll model as used in BJT. Also derive the expression of $V_{CE}(sat)$ from this model. 9

(b) Explain high level injection and current crowding effects in a BJT with necessary figures. How can the problem of current crowding be solved? 8

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(c) Consider a silicon n-channel JFET at $T = 300$ K with the following parameters: $N_a = 1 \times 10^{18}$ cm^{-3}, $N_d = 1 \times 10^{16}$ cm^{-3}, $a = 0.75$ μm, $L = 10$ μm, $W = 30$ μm, and $\mu_n = 1000$ cm^2/(V s)

(i) Calculate the maximum current in an n-channel JFET.

(ii) Determine the maximum transconductance of an n-channel depletion mode JFET biased in the saturation region. $I_{P1} = 0.522$ mA, $V_{bi} = 0.814$ V, and $V_{po} = 4.35$ V. 8

Term Test 1 (its questions 1 to 4 are 7(a) to 7(d))

[TT1, EEE 459; Marks: 20; Time: 30 minutes]

7. (a) Why is the electron mobility of Si less than GaAs? 3

(b) What is Gunn effect? 3

(c) Prove that along the [100] and [111] direction in k-space, for Si, the variation of electron energy with respect to wave number is higher for light holes than that of heavy holes. 4

(d) For a GaAs-$Al_xGa_{(1-x)}As$ heterojunction, determine the following:

(a) Energy-Band Diagram: (i) before contact (ii) After contact

(b) Show that the built-in potential (using hole concentration) $eV_{bi} = -\Delta E_v + kT \ln\left(\frac{P_p N_{VN}}{P_N N_{VP}}\right)$. Given, GaAs is doped p type, $Al_xGa_{(1-x)}As$ is doped N+ type. Assume: x=0.25 & $\Delta E_c = \frac{1}{2}\Delta E_g$. P_p = Hole concentration in P type region, P_N = hole concentration in N type region, N_{VN} and N_{VP} are density of states of electron and hole, respectively. *Hole concentration*, $P = N_V \exp\left(\frac{-(E_F - E_V)}{kT}\right)$ 10

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